

Candidate of **CFA level 1**, Aug 2024 | Secured **99.3 %ile** in JEE MAINS | Ranked **61** in Maharashtra CET exam, **100%ile** in Maths

PROFESSIONAL EXPERIENCE

Maritime Research centre Research Intern [May'24 - Jul'24]	Awarded LOR from the Director Research on Statistical Inference of climate change impact on India's GDP	
	Research	- Diligently studied 10+ research paper on climate change in IOR and impact on India's GDP - Analyzed sea temperature data with Python correlating risk to underwater ecosystem - Worked on implementing LSTM based ML model for tropical ocean SST forecasting
	Publication	- Authored two research notes and articles followed by presenting at two ESG conferences - Article on climate change risk assessment in the Indian Ocean region published publicly
DRDO AI/ML Research Internship [Dec'23-Jan'24]	Received a LOR for hard work & diligence Worked on drone navigation system using optical flow sensors	
	Drone navigation	- Critiqued the topic Optical flow estimation for drone navigation in GPS denied environment - Developed a simulation for numerical based & CNN based algorithm to track object motion
	JetSon	Implemented trained model on NVIDIA JetSon with optical flow sensor for real time tracking

POSITION OF RESPONSIBILITY

NSS IITB CTM [May'23 - Apr'24]	National Service Scheme, largest student-volunteer body of the nation serving 1 million+ people nationwide	
	Management	- Led 2 tier team of 120+ volunteers & 9 coordinators to reduce carbon footprint in campus - Incorporated 30+ small SMEs for a social entrepreneurship event promoting inclusivity
	Initiatives	- Initiated development of a self dripping system, reducing 10+ working hours per week - Organized tulsi plantation drive of 2000 sapling reducing almost 10+ ton CO₂ per year
ISMP Mentor , SMP IITB [Jul' 24-Present]	SMP IIT Bombay, student driven body, providing help to students through constant support and mentorship - Selected among 150 out of 435 applicants through a process involving SoP, peer reviews, and interviews - Enriching undergraduate fresher's campus transition issues through continuous mentorship and support.	

KEY PROJECTS & COMPETITIONS

Quantitative Finance & Trading Strategies [WorldQuant]	Globally among 3K consultant out of 0.1 M+ users, ideating algorithms for market above average return	
		- Scrutinizing ideas to build alpha on market securities with subsequent backtesting on real financial data - Ideated strategies on MA CrossOver, subsequently achieving 25%+ returns on quarterly time frame data - Developed robust alpha expressions with sharpe ratio above 2, demonstrating high-risk adjusted return.
Option Pricing model & accuracy [Finsearch]	Researched on European call options pricing using Black- Scholes model Participation of 500+ students - Back-tested Black Scholes Option Pricing Model on Nifty50 ATM call option, achieved RMSE value 2.25 - Understood the working of option pricing models and option greeks to find optimal price of derivatives	
Sentimental analysis [Self-Project]	Classified IMDb movies reviews based on the movie ratings and genre to quantify subjectivity and polarity Web scraping: Preprocessed & organized movie data in a tabular form using BeautifulSoup Python library - Implemented TextBlob Python library defining subjectivity & polarity of reviews, ultimately the genuinity	
Monte Carlo Simulation model [Seminar Course]	Guide: Professor M.P Gururajan Mathematical modelling & simulation of material microstructure evolution - Implemented Monte Carlo simulation technique to model spin interaction among atom in binary solution - Analyzed microstructure evolution of material and programmed Metropolis method simulation in Python	

EXTRACURRICULARS

AWARDS & VOLUNTEERING	- Recipient of academic excellence award for 3 consecutive years from Coast Guard, Ministry of Defence NCC cadet: completed 80+ hours training Activity Associate NSS: Ideated activities for underprivileged
MENTORSHIP	SOC Mentor: Upskilling 10+ students, providing constant guidance on CNN based face-detection project
CERTIFICATES	Machine learning: Regression & classification, Advanced learning algorithm Finance: Excel fundamentals
MISCELLANEOUS	Managing my own investment portfolio of INR 0.1 M, generated 15% return in single quarter Y-O-Y basis Ranked in top 2% all over the world in Clash Royale Esports gaming among 10 million+ users worldwide

